

Image Classification with Neural Networks Using the Fruits Dataset

Abstract

Image classification is one of the important tasks in computer vision. In this study, a Convolutional Neural Network (CNN) was used for image classification of fruits using Fruits Image Dataset available on Kaggle. Before training the network, pre-processing of images was done by resizing and normalizing them. The proposed CNN network structure includes convolutional layers, pooling layers, dropout layers, and dense layers. The training of the network was done using the Adam optimization algorithm, and its performance was evaluated by using classification accuracy and analysing the confusion matrix. The results showed that CNN models can learn important features from images and perform classification of images for different classes of fruits.

Introduction

Image classification is a technique that is used in a wide range of applications, such as agriculture, food recognition, and inspection systems. Recently, deep learning techniques, especially Convolutional Neural Networks, have improved the accuracy of image recognition tasks significantly. Convolutional Neural Networks can learn significant features from images, such as edges, shapes, and texture, automatically. The main aim of this study is to develop a Convolutional Neural Network-based model for accurate recognition of various types of fruit images.

Dataset

The Fruits Image Dataset downloaded from Kaggle was used in this research. The dataset contains images of different classes of fruits, where each class is stored in its own folder. Before the model was trained, all images were resized to 100 x 100 pixels, and the pixel values were normalized to the range 0 to 1 from the original 0 to 255. In order to effectively test the model, the images were randomly split into subsets.

Fruit Classes: Apple, Grapes, Orange, Pineapple, Strawberry,
Training Images: 640
Validation Images: 160
Dataset Link:

<https://www.kaggle.com/datasets/ayessa/fruits-image-dataset>

Model Architecture

The A Convolutional Neural Network (CNN) model was developed using multiple layers, which include Conv2D layers, MaxPooling layers, a Flatten layer, Dense layers, and a Dropout layer to avoid overfitting in the network. The network takes RGB images of size 100 x 100 x 3 and extracts hierarchical features from the images. These features are then used in the final classification of the images.

Training

In this regard, the training of the model was performed using the Adam optimization algorithm coupled with a sparse categorical cross-entropy loss function. The batch size for this training was 32, while 10 epochs were used for training. The training accuracy and validation accuracy

were monitored during training for the evaluation of the training performance of the model.

Results

The CNN model achieved satisfactory results in terms of classification accuracy. In order to evaluate the prediction performance, metrics like confusion matrices and classification reports were used. It was evaluated that the model misclassified the images in a few cases, especially between similar-looking fruits.

Final Test Accuracy: 83.5%

Test Loss: 0.4562

Conclusion

Thus, this study demonstrates that it is possible for a Convolutional Neural Network (CNN) model to classify images of fruits with a high degree of precision. Future research could focus on enhancing this model by using deeper network architectures, incorporating data augmentation methods, and using large datasets for training in order to improve classification precision.

References

- [1] Kaggle, Fruits Image Dataset.
<https://www.kaggle.com/datasets/ayessa/fruits-image-dataset>
- [2] I. Goodfellow et al., Deep Learning, MIT Press.